DOC 314

Tutorial

Task 2

A) Generations of computers

1) First generation

Computer development's early stage is referred to as the first generation of computers. This generation signifies the beginning of computer technology, which has advanced to produce computers more powerful and smaller.
The first generation of computers are considered as functioning from the year's 1946 to 1959, despite the fact that their development started around 1940. Vacuum tubes were the primary component used in the development of computers during this time. Circuitry for the central processing unit was made possible in particular by vacuum tubes. Punch cards, paper tape, and magnetic tape were some of the additional fundamental elements used for input-output devices, as well as for memory requirements.

Due to the massive size of the vacuum tubes the computer appeared to be gigantic as well and took up a lot of space in a room. Additional to that , vacuum tubes consumed a large amount of electricity and during the installation it often fused due to its complex nature. All these factors resulted in making the first generation of computers immensely expensive and making it affordable only for large organizations.
When it comes to software needs, first generation computers relied on machine language, known as the lowest level programming language and easily understood by computers, for its software requirements. The ability solve one operation at a time allows computers to attend a range of jobs but it didn't have the ability of multitasking. In order to finish an operation, the computers even took days or weeks.

Some of the examples for the first-generation computer are,
ENIAC (Electronic Number Integrated and Calculator) ,
UNIVAC (UNIVersal Automatic Computer)

2) Second generation

Even though the development of computers commenced in the first generation, but there were some issues related to its size, speed, reliability and etc. All these problems resulted in the introduction of the second generation of computers.

Second generation of computers were developed using transistors as the basic component instead of vacuum tubes which were used in the first generation. Due to the relatively small size, fast in terms of speed, and cost effectiveness the transistors were considered to be much effective that the vacuum tubes. Also, it consumed less amount of energy and allowed the computer to be more reliable.

Unlike, the first generation of computers, assembly language was used instead of binary machine language in the second generation computers. This allowed developers to input instructions in words. Also, high level languages such as COBOL and FORTRAN were introduced as the early versions which were also used in the second generation computers. They typically used batch processing combined with the multiprogramming operation system.

Second generation computers were widely used for commercial purposes, especially for businesses and scientific jobs. UNIVAC 1108, CDC 3600, IBM 7070, and IBM 7094 are some remarkable examples for the second generation computers.

3) Third generation

With the third generation of computers, innovation and computer development were further advanced. The development of third generations of computers were marked by the period when the transistors were replaced by integrated circuits. These were widely used as the primary technology in third generation computers. in particular, the transistors were assembled on silicon chips called, semiconductors, which resulted in the computer and improved efficiency.

In the third generation, computers had better input-output devices. The use of keyboards and monitors was introduced in place of punched cards and printouts, which helped in increasing the speed of input and output operations. Besides, remote processing, time sharing and multiprogramming operating systems were also introduced in this generation, which eventually allowed the users to run multiple applications at once.

Some of the examples for the third generation of computers include, IBM 370/168, PDP 11, TDC 11 and etc.

4) Fourth generation

the fourth generation of computers was a progressive version of the third generation of computers. This eventually added more benefits to computer technology. Fourth generation of computers began with the use of microprocessors in computer systems. Due to hundred of thousands of integrated circuits could be assembled on a single silicon chip this is considered as a revolution in the computer industry. Also, this enabled the production of compact sizes of computers which can easily fit in a desk.

Apart from the size fourth generation computers were also more powerful, reliable, and inexpensive. This eventually led to the widespread use of personal computers (PCs). This means that computer systems became accessible to mass audience due to their portability. Besides time-sharing, time networking based, distributed operating systems were used in computers. When it comes to languages, fourth generation computers used high level programming languages such as C, C++, DBASE, and more.

IBM 4341, PDP 11, DEC 10 and STAR 1000 are some of the examples for fourth generation computers. In addition to these super computers such as, CRAY-1 and Cray-X-MP also will be considered as examples.

5) Fifth generation of computers

The fifth generations of computers are by far the most successful generation of computers. In this generation, there have been significant changes in significant in computer technology that make computers extraordinary. Instead of VLSI technology used in fourth generation computers, fifth-generation computers received a major upgrade and VLSI technology was replaced with ULSI (Ultra Large-Scale Integration) technology. In addition to this technology, fifth generation computers were also based on AI (artificial intelligence), which is still in the development stage, and parallel processing hardware. Artificial intelligence is an emerging technology in computer science that primarily helps to make computers more realistic. This means that it enables computers to think like humans; however, there is much more room for improvement as well.

Fifth generation computers are incorporated with logic programming and massively parallel computing. In terms of programming languages, all high level languages are used in fifth generation computers. Some of the most common languages are C, C++, Java, .Net, etc.

There are numerous potential applications of artificial intelligence that are included in fifth generation devices. Some common examples of fifth generation computers are,

Desktop, Laptop, Notebook, ChromeBook

B) Computer

Definition:

A computer is a device that performs processes, calculations and operations based on instructions provided by a software program, which has the ability to accept data (input), process it, and then produce outputs.

Components of a computer

A computer device is made up of various elements which help in its effective functioning and processing. There are five basic components of the computer which makes the processing of data convenient.

By definition, components of a computer system are the primary elements which makes the functioning of an electronic device smooth and faster. The components include,

1. Input unit
2. Output unit
3. Memory unit
4. Control unit
5. Arithmetical and logical unit

- **Input unit**

A computer will only respond when a command is given to the device. The input or the input device can be used to issue these commands. For instance, we can type a text into a notepad using a keyboard, and the computer will process the data and display the results on the screen. The data entered may appear in forms of letters, numbers, or graphics, among other things. We enter data via an input device, the processing units translate it into computer understandable languages, and finally a human understandable language is obtained as the output.

- **Output unit**

When we command a computer to perform a certain task, it reverts for the action performs and gives us an outcome. This result is known as the output. The computer is connected to a variety of output devices. A monitor is among the most elementary. Everything we type on the keyboard or click with the mouse is seen on the monitor.

Thus, after all processing is completed inside a device's mechanism, the output unit provides us with the final outcome.

- **Memory unit**

Using an input device, we can enter data into a computer, the entered information immediately gets saved in the memory unit of the Central processing unit (CPU). Due to the presence of a certain programming, the memory unit sends the information on to the other CPU components.

Similar to this, before the output is given to the user after the computer has processed our command, it is saved in the memory unit.

- **Control unit**

This is the core unit which controls how the computer system operates as a whole. One of the most important parts of the computer system. The data entered through the input unit is collected by the control unit, which then sends it for processing, receives the result, and displays it to the user. It can be referred to as the hub of all processing operations occurring within a computer device.

In essence, the control unit handles all the instructions received, data interpretation, signaling for data execution, and data retrieval.

- **Arithmetical and logical unit**

All the mathematical calculations or arithmetical operations are performed in the arithmetical and logical unit of the CPU. It can also perform actions like a comparison of data and decision-making actions. The ALU comprises circuits using which addition, subtraction, multiplication, division and other numerical based calculations can be performed.